# Automated Water Quality & Contamination Detection Using Computer Vision, Machine Learning, Deep Learning and Generative AI

## INDIVIDUAL THESIS

R.M.N.A. RATHNAYAKE

(IT22193940)

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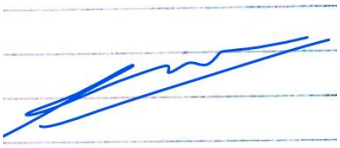
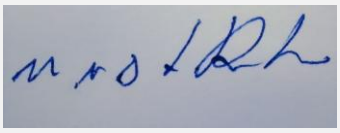
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Student name: R.M.N.A. RATHNAYAKE

Student ID: IT22193940

Signature:  Date: __11/10/2026__

**Supervisor certification.** The above candidate carried out this research for the bachelor's degree thesis under my supervision.

| Role and name | Signature | Date |
|---|---|---|
| Supervisor: Mr. Buddika Harshanath |  | 11/10/2026 |
| Co-supervisor: Prof. Samantha Rajapaksha |  | 11/10/2026 |

# ABSTRACT

This thesis draft develops the proposed component described in the supplied project documents. Implementation, experiments, participant studies and measured results have not been supplied; corresponding chapters therefore state the planned work and evidence required.

The Automated Water Quality and Contamination Detection project combines image-based contamination detection, physicochemical assessment and microscopic analysis. This individual component proposes a Water Safety Report and Purification Advisor that converts the outputs of those modules into understandable, evidence-linked reports. The problem addressed is the separation between technical detection results and guidance that users can interpret without losing uncertainty, provenance or treatment constraints. The proposed approach combines retrieval-augmented generation with a curated water-quality knowledge base, structured input validation and deterministic treatment calculations. A rule layer will determine sufficiency and permitted recommendations before a language model explains the findings with citations. A comparative study is planned against a template-based report, an ungrounded language model and a basic retrieval-augmented baseline. Evaluation will measure retrieval relevance, factual support, citation correctness, recommendation appropriateness, abstention, numerical correctness and user comprehension. No empirical results are reported because a working implementation and evaluation dataset were not provided. The intended individual contribution is an auditable report-generation and advisory pipeline, together with an evaluation protocol for assessing its reliability. The component is designed as decision support, with explicit handling of missing or contradictory evidence, rather than as an independent certification of drinking-water safety.

Keywords: water quality, retrieval-augmented generation, generative AI, decision support, purification advice

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# LIST OF ABBREVIATIONS

Abbreviations used in this document are listed below.

| Abbreviation | Meaning |
|---|---|
| AI | Artificial intelligence |
| API | Application programming interface |
| GenAI | Generative artificial intelligence |
| JSON | JavaScript Object Notation |
| LLM | Large language model |
| RAG | Retrieval-augmented generation |
| SDG | Sustainable Development Goal |
| TDS | Total dissolved solids |
| WHO | World Health Organization |

# 1. INTRODUCTION

## 1.1 Overall project context

The overall research project, **"Automated Water Quality and Contamination Detection Using Computer Vision, Machine Learning, Deep Learning and Generative AI,"** proposes an integrated system for assessing water samples using multiple sources of evidence. Its purpose is to combine analytical findings into an understandable assessment that supports water-quality monitoring and informed decision-making.

The system consists of four interconnected components. Contamination Detection analyses water photographs to locate visible contamination indicators, such as floating debris, discoloration, oil sheen and algae-like regions. Water Quality Classification processes measured physicochemical parameters, including pH, hardness, turbidity, total dissolved solids and fluoride, to predict a defined quality category and identify parameter-level concerns. Microorganism Detection examines microscope images to locate microorganisms and predict supported species or genus labels. Finally, the Water Safety Report and Purification Advisor combines these outputs and uses generative AI with retrieval-augmented generation (RAG) to produce explanations and relevant purification guidance grounded in trusted reference material.

Within this overall framework, the individual research component focuses on the reporting and advisory stage. It aims to translate technical findings into clear, traceable reports while preserving uncertainty, identifying missing or conflicting evidence, and explaining the limitations of the assessment. The proposed system is intended to support supervised water-quality assessment; its effectiveness must be established through implementation and validation.

## 1.2 Individual component context

The Water Safety Report and Purification Advisor will receive structured sample findings and produce a readable assessment containing evidence, limitations, source references and relevant next steps. Generative artificial intelligence (GenAI) will be used for explanation. Retrieval-augmented generation (RAG) will provide the selected reference passages. Explicit rules and calculation functions will control decisions that should not depend on free-form generated text.

The intended users are water-quality practitioners, community monitoring personnel and non-specialist users viewing the group dashboard. The initial prototype will support English reports for supervised research use. A single report will distinguish measured observations, upstream model predictions and reference-based advice.

### 1.3 Problem statement

Outputs such as contamination labels, prediction probabilities and numerical measurements do not themselves explain what a user should do. Combining them carelessly can also hide disagreement: a favourable classifier output may coexist with a microbial warning or absent measurements. A generic language model may produce fluent advice without adequately connecting it to the sample evidence.

The research problem is how to generate a clear, traceable and appropriately cautious advisory report from heterogeneous and potentially incomplete water-quality findings. The system must explain what is known, preserve uncertainty, identify missing information and avoid generating treatment quantities without validated inputs.

### 1.4 Research gap

The research gap addressed by this study concerns the translation of heterogeneous water-quality findings into clear, traceable and context-appropriate purification guidance. Integrating photographic observations, physicochemical classifications and microorganism predictions requires a reporting framework that preserves uncertainty, identifies missing information and explicitly addresses conflicting evidence.

Although retrieval-augmented generation (RAG) provides a foundation for grounding explanations in external reference material [1], retrieval alone does not establish whether a recommendation is appropriate for a particular sample, whether its prerequisites are satisfied or whether associated numerical calculations are valid. The proposed study therefore investigates a hybrid framework combining RAG with structured evidence validation, explicit decision rules and deterministic calculations.

The framework will be evaluated against template-based reporting, an ungrounded language model and basic RAG to determine whether these additional controls improve factual support, recommendation appropriateness and user understanding. The intended contribution is an empirical assessment of these controls within water-quality reporting; broader claims of novelty remain dependent on a comprehensive review of comparable advisory systems.

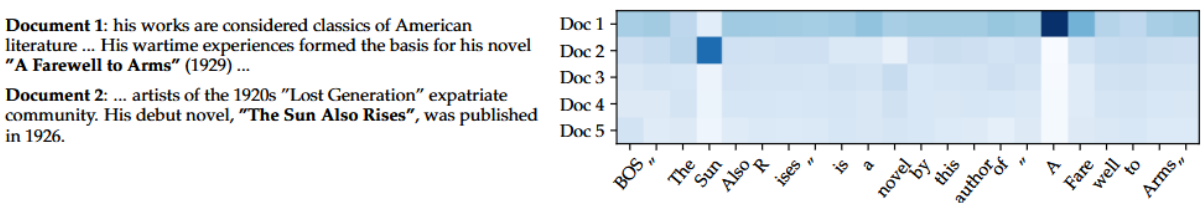



**Figure 1: RAG-Token document posterior**

### 1.5 Aim and objectives

Aim: To design and evaluate an evidence-grounded Water Safety Report and Purification Advisor that transforms integrated water-quality findings into understandable reports with traceable, context-appropriate guidance.

Objectives:

1: Specify a common input schema and a report schema that preserve sample identity, units, provenance and uncertainty.
2: Curate and version a reference corpus, and implement a retriever that returns relevant, attributable passages.
3: Develop a hybrid reporting pipeline combining evidence-sufficiency rules, RAG and separately validated calculations.
4: Integrate the component with the three upstream modules and test incomplete, conflicting and invalid inputs.
5: Evaluate factual support, recommendation appropriateness, citation quality, abstention, numerical accuracy, operational performance and user comprehension against defined baselines.

### 1.6 Research questions or hypotheses

Q1: Does the proposed hybrid pipeline reduce unsupported claims and inappropriate recommendations compared with an ungrounded language model and a basic RAG pipeline?
Q2: How do retrieval choices affect the relevance and citation coverage of reports generated for different contamination scenarios?
Q3: Does explicit evidence validation and abstention improve handling of missing, contradictory or low-confidence findings?
Q4: Can deterministic calculation functions preserve numerical correctness while generated explanations remain understandable?
Q5: How do users interpret the report's evidence, limitations and recommended next steps compared with a fixed reporting template?

### 1.7 Scope limitations and assumptions

The component will consume upstream outputs; it will not train the contamination, physicochemical or microorganism classifiers. Its scope includes report generation, source retrieval, evidence checks, treatment-option matching and calculation support for explicitly approved procedures. It excludes autonomous dosing equipment, clinical diagnosis and certification that a water supply is potable.

The design assumes access to permitted reference material and qualified reviewers. Model confidence will be treated as a property of a prediction, not as a probability that water is safe. Missing measurements will remain missing rather than being inferred by the language model. A prototype using synthetic cases can establish software behavior but cannot establish real-world treatment effectiveness. Applicable local requirements must be identified before operational deployment.

### 1.8 Individual contribution

The proposed individual work comprises the evidence schema, knowledge-based preparation, retrieval configuration, risk and abstention rules, controlled calculation interface, report prompt and output validator, dashboard/API integration and evaluation framework. Contributions from other team members will be acknowledged through interface ownership and input provenance. Actual implementation ownership should later be supported by repository history, design records and test artefacts.

### 1.9 Thesis structure

This thesis is organized into eight chapters. Chapter 1 introduces the overall project, individual component, research problem, gap and objectives. Chapter 2 reviews relevant literature on water-quality guidance, generative AI and RAG. Chapter 3 presents the research methodology, data requirements and evaluation procedures. Chapter 4 describes the proposed system architecture and component design. Chapter 5 outlines implementation, integration and testing. Chapter 6 is reserved for experimental results and validation findings once the evaluation is completed. Chapter 7 discusses the findings, contributions and limitations, while Chapter 8 concludes the study and identifies future research directions. References and appendices provide supporting sources, research instruments and supplementary documentation.

**Table 1: Examples from generation tasks. RAG models generate more specific and factually accurate**

| Task | Input | Model | Generation |
|---|---|---|---|
| MS-MARCO | define middle ear | BART | ?The middle ear is the part of the ear between the middle ear and the nose. |
| | | RAG-T | The middle ear is the portion of the ear internal to the eardrum. |
| | | RAG-S | The middle ear includes the tympanic cavity and the three ossicles. |
| | what currency needed in scotland | BART | The currency needed in Scotland is Pound sterling. |
| | | RAG-T | Pound is the currency needed in Scotland. |
| | | RAG-S | The currency needed in Scotland is the pound sterling. |
| Jeopardy Question Gener -ation | Washington | BART | ?This state has the largest number of counties in the U.S. |
| | | RAG-T | It's the only U.S. state named for a U.S. president |
| | | RAG-S | It's the state where you'll find Mount Rainier National Park |
| | The Divine Comedy | BART | *This epic poem by Dante is divided into 3 parts: the Inferno, the Purgatorio & the Purgatorio |
| | | RAG-T | Dante's "Inferno" is the first part of this epic poem |
| | | RAG-S | This 14th century work is divided into 3 sections: "Inferno", "Purgatorio" & "Paradiso" |

# 2. LITERATURE REVIEW AND RELATED WORK

## 2.1 Review method and scope

This study uses a focused, non-systematic literature review of peer-reviewed research, official water-quality guidelines and relevant technical documentation. The review covers retrieval-augmented generation, evidence-grounded report generation, purification guidance, deterministic calculations and evaluation of recommendation reliability. Sources are considered according to their relevance, authority, methodological clarity and applicability to the proposed component. Attention is given to citation accuracy, uncertainty handling and the suitability of recommendations for specific water-quality findings. The review focuses on the Water Safety Report and Purification Advisor; upstream contamination detection and classification methods are discussed only where they affect integration and interpretation.

## 2.2 Domain and theoretical foundations

Water-quality assessment and treatment guidance require attention to the conditions under which evidence applies. WHO's current fourth edition incorporating three addenda presents risk management across the drinking-water supply chain and supports health-based targets and surveillance [2]. In this project, authoritative guidance will inform a versioned corpus; numerical criteria and local applicability will require explicit expert verification before being encoded. The WHO Water Safety Plan Manual provides a practical framework for risk assessment and management [3]. WHO guidance on household water treatment provides microbiological performance criteria for evaluating treatment options; its scope does not establish removal of every chemical contaminant [4].

A large language model (LLM) will act as the explanatory layer. RAG supplies external passages relevant to a query [1], while the project's proposed rule layer retains control over evidence sufficiency and allowed actions. Retrieval relevance, correct interpretation and source applicability are distinct properties: finding a related passage does not establish that its procedure is suitable for a particular sample.

## 2.3 Related research

Related research has explored RAG models that combine language generation with external knowledge retrieval to support evidence-grounded explanations [1]. Evaluation frameworks assess retrieval quality, context relevance, answer faithfulness and answer relevance, highlighting the importance of evaluating retrieval and generation separately [5], [6]. However, these

approaches do not establish the accuracy of water-treatment advice. Therefore, this study will supplement automated evaluation with qualified human review to assess recommendation appropriateness and identify harmful omissions, rather than treating fluent language or the presence of citations as sufficient evidence of reliability. Research on citation-supported generation further motivates assessing whether citations substantiate the associated claims [7]. Dense Passage Retrieval provides a foundation for retrieving relevant passages using learned representations [8], while Sentence-BERT provides a foundation for sentence embeddings and semantic similarity [9]. These methods inform candidate retrieval designs rather than demonstrate water-treatment accuracy.

### 2.4 Existing systems products and techniques

Existing approaches relevant to this component include template-based reporting, general-purpose language models and retrieval-augmented generation (RAG). Template-based systems present findings consistently but offer limited flexibility in explaining complex cases. General-purpose language models generate readable explanations, although their recommendations may lack supporting evidence. RAG systems retrieve reference material to support responses [1], but retrieval alone does not ensure that advice is suitable for a particular water sample. The proposed component therefore combines RAG with structured input validation, explicit decision rules and deterministic calculations to produce traceable reports and context-appropriate purification guidance.

### 2.5 Gap synthesis and conceptual framework

The identified research gap is the need to combine diverse water-quality findings with evidence-grounded guidance that accounts for uncertainty, missing information and treatment requirements. The proposed conceptual framework links structured input validation, knowledge retrieval, decision rules and deterministic calculations to report generation. Findings from the three analytical components are first checked for completeness and consistency, then relevant guidance is retrieved and assessed for applicability. The language model explains the supported findings and recommendations, while output validation checks citations and numerical consistency. The study will evaluate whether this combined approach improves factual support, recommendation appropriateness and user understanding compared with simpler reporting methods.

# 3. RESEARCH METHODOLOGY

## 3.1 Research design

This study adopts a design-science research approach to develop and evaluate the Water Safety Report and Purification Advisor [10]. The process includes identifying reporting requirements, designing the architecture, developing a prototype and evaluating it using reviewed water-quality scenarios. The proposed approach will be compared with template-based reporting, an ungrounded language model and basic RAG. Evaluation will assess factual support, recommendation appropriateness, citation correctness, numerical accuracy and handling of incomplete or conflicting evidence. Expert review and a small user study will also assess reliability and comprehension. Development will be iterative, while separate held-out cases will be reserved for the final evaluation.

## 3.2 Data participants or study materials

Three types of material are required: an approved reference corpus; structured sample cases; and reference judgements for evaluating those cases. Corpus records will include publisher, title, edition/date, URL, section or page, licence status, extraction date and reviewer approval. The initial source family is WHO guidance [2]-[4], supplemented only by reviewed treatment references and applicable local material when obtained.

A feasible initial target is 120 independently reviewed scenarios: 40 development cases and 80 held-out evaluation cases. These numbers are planning assumptions, not collected data or power calculation. Cases should cover physicochemical concerns, microbial flags, visible contamination, mixed hazards, apparently normal measurements, missing fields, conflicting findings and unsupported requests. Related variants from the same base scenario must remain in one split.

Two qualified reviewers are proposed for independent reference judgements, subject to availability. A formative comprehension study with approximately 12–20 consenting target users is proposed, subject to institutional approval. Synthetic scenarios must be labelled and reported separately from any real, authorized samples.

## 3.3 Experimental or study procedure

First, agree with the schema and reviewer rubric, curate the corpus, and construct development cases. Next, annotate relevant passages, acceptable report conclusions, required limitations and prohibited recommendations. Configure retrieval and prompts using only the development split.

Freeze the corpus snapshot, model identifier, prompts and rule version before evaluating held-out cases.

Compare four configurations: B0, a deterministic template using the common validated inputs; B1, an LLM using the same structured inputs without retrieval; B2, the same LLM with basic RAG; and P, the proposed RAG pipeline with evidence checks, action constraints and deterministic calculations. Use the same model and generation settings for B1, B2 and P. All configurations receive the same cases. Unsafe experimental outputs remain in a restricted evaluation environment.

Within development data, compare keyword and vector retrieval with a hybrid option; test chunk lengths of approximately 300 and 600 tokens and retrieval depths of 3 and 5 as candidate settings. Preserve tables and procedural conditions even when they cross these nominal lengths. Select settings before the final run. Repeat generative configurations three times per held-out case to observe output variability; case-level analysis must account for repeated outputs.

### 3.4 Measures and evaluation metrics

Table 3.1 defines the planned measures. Human annotation is the primary reference for advice appropriateness. Retrieval will be measured only in cases with a judged relevant evidence set. Results must report denominators, exclusions and missing scores; a report with no substantive claims must not receive perfect factual-support credit merely by abstaining. The separation of retrieval relevance, faithful generation and citation support is informed by RAGAs, ARES and citation-generation research [5]– [7].

Abstention will be assessed in both directions: correctly withholding unsupported advice and unnecessarily withholding an answer when evidence is sufficient. Report coverage is the proportion of answerable cases receiving an adequate substantive response. Calculation tests will use independently checked reference values and tolerances defined per function.

**Table 2: Evaluation measures**

| Measure | Operational definition |
|---|---|
| Retrieval recall@k | Relevant reference passages retrieved / all annotated relevant passages, per case. |
| Citation correctness | Cited claims adequately supported by their cited passage / cited claims assessed. |
| Citation coverage | External factual claims with adequate supporting citations / claims requiring a citation. |

| Measure | Operational definition |
|---|---|
| Unsupported-claim rate | Unsupported substantive factual claims / substantive factual claims assessed. |
| Inappropriate-advice rate | Reports containing at least one reviewer-labelled inappropriate action / reports reviewed. |
| Abstention and coverage | Report corrects abstention on unanswerable cases and adequate answers on answerable cases separately. |
| Calculation correctness | Cases meeting independently checked values and units / eligible calculation test cases. |
| Comprehension and operation | Task accuracy and time; median and 95th-percentile report latency; cost per report. |

### 3.5 Data analysis

Compute per-case scores and paired differences between configurations. Report uncertainty using a case-level bootstrap, preserving repeated outputs from the same case as a cluster. For binary advice outcomes, a paired categorical analysis may be used when its assumptions and counts are suitable. Any inferential analysis will be specified before inspection of held-out outcomes.

Report raw reviewer agreement and an appropriate chance-adjusted agreement measure before resolving disagreements. Present outcomes by scenario category, with particular attention to false reassurance and missed warnings. User-study outcomes will include task accuracy, completion time and thematic feedback; a small convenience sample will support formative findings rather than population-wide conclusions.

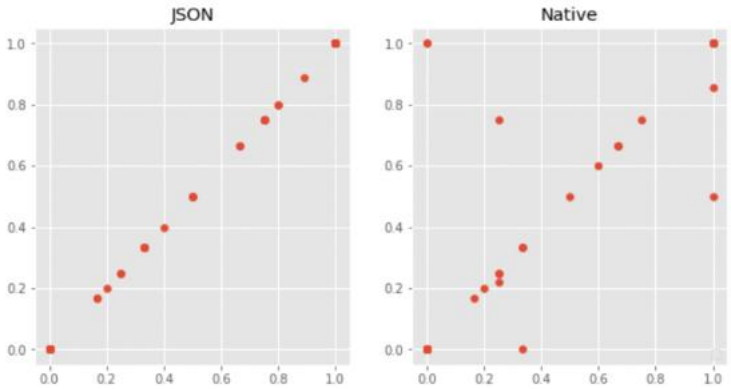



**Figure 2: Consistency of RAGAs scores across two different runs of the model**

### 3.6 Ethics privacy security and responsible AI

Obtain required institutional approval before recruiting participants or collecting identifiable field data. Participants will receive an explanation of the research prototype, consent procedures and withdrawal options. Sample records will use pseudonymous identifiers; unnecessary names and exact household locations will not be sent to external model services.

Access control will separate users, reviewers and corpus administrators. Treat retrieved text as data rather than executable instructions and restrict the model to approved tools and sources. Corpus ingestion will include review for irrelevant or malicious instructions. Reports must distinguish predictions from verified observations and make insufficient evidence visible. No prototype output will be used to administer treatment in this study.

### 3.7 Reproducibility and research artefacts

Retain the source-code revision, dependency lock file, model and embedding identifiers, generation parameters, prompt text, retrieval settings, corpus manifest, chunk IDs and rule versions. Each evaluation record will contain case ID, split, retrieved passages, generated output, validator outcomes, latency, token usage and reviewer labels.

Publish or submit only data permitted by licences and consent. Where raw material cannot be redistributed, provide metadata and reconstruction instructions. Appendix A specifies the evaluation instrument and representative tests. Appendix B records the drafting assistance used for this document.

# 4. PRODUCT AND SOLUTION DESIGN

## 4.1 Requirements

Functional requirements are to accept validated multi-module findings; preserve sample identity and provenance; retrieve approved reference passages; identify conflicting or insufficient evidence; generate a structured, readable report with citations; return relevant treatment options only when conditions are satisfied; calculate quantities through approved functions; and export report and audit metadata.

Non-functional requirements are traceability, predictable failure handling, access control, maintainability and understandable presentation. Operational latency and cost targets will be established after a prototype baseline. A release gate should require all predefined critical safety tests and calculation tests to pass; this does not imply zero residual risk in unseen situations.

## 4.2 Component architecture within the complete solution

The reporting service sits downstream of the three analytical components and upstream of the dashboard. The proposed processing sequence is input validation, sample-level evidence aggregation, rule evaluation, retrieval, option eligibility checks, optional calculation, constrained generation, output validation and report delivery. Table 4.1 defines the integration boundaries.

The LLM will receive approved findings and allowed recommendation objects. It will not decide which sample records to merge or silently override rule outcomes. A shared audit record will connect each report to the input snapshot, corpus version and model configuration. If a dependency fails, the service will return to a structured partial result or an explicit unavailable status.

**Table 3: Component interfaces**

| Boundary | Required information | Reporting responsibility |
|---|---|---|
| Visible contamination module | Sample ID, labels, confidence, acquisition metadata | Retain as predicted visual findings. |
| Physicochemical module | Measurements, units, quality label, method | Show evidence and check required fields. |
| Microorganism module | Findings, counts were meaningful, confidence, method | Preserve suspected/confirmed distinction. |
| Knowledge service | Approved passages, source IDs, versions | Attach provenance and applicability. |

| Boundary | Required information | Reporting responsibility |
|---|---|---|
| Calculation service | Approved procedure ID, validated inputs and units | Return checked values or blocked status. |
| Dashboard | Report JSON, citations, limitations, audit ID | Display scope and uncertainty with the report. |

### 4.3 Design alternatives and decisions

RAG is selected for its ability to attach updateable reference material to a report [1]. Fine-tuning alone is not proposed as the source of current guidance because the study needs inspectable source of provenance. A deterministic template remains a baseline and a fallback for validated observations when generation fails.

The proposed hybrid design separates semantic retrieval from numerical calculation. A vector index supports meaning-based retrieval [8], [9], while keyword matching can help preserve exact parameter names and identifiers. The final retrieval method will be selected experimentally. The design avoids averaging upstream confidence scores because they originate from different models and may not share a calibration scale.

### 4.4 Detailed design

The input object will include sample_id, schema_version, collection time, intended use, parameter observations with values and units, detected contamination labels, microbial findings, confidence metadata, method identifiers and missing-field flags. Reports will include assessment_status, evidence_summary, uncertainties, advice_items, missing_information, citations and audit_id.

A proposed status vocabulary is concern_detected, insufficient_evidence and no_concern_in_assessed_scope. The last status will never be displayed as a blanket declaration of potability. If evidence conflicts, the report will describe the conflict and request review; absent microbial results will not be interpreted as a negative microbial test.

Reference documents will be divided by meaningful sections, retaining headings, tables and conditions. Each chunk will have a stable ID and version. Retrieval will filter by source approval, intended use and applicable context before ranking. Advice items will reference their supporting chunk IDs and prerequisite conditions. Retrieved text will be delimited so it cannot redefine the system’s instructions.

A calculation function will run only after an approved procedure has supplied the required treatment specification. For software unit testing, the dimensional relationship $m = C \times V$ relates

active mass m in milligrams to specified concentration C in milligrams per litre and volume V in litres. If a verified stock concentration S is expressed in milligrams per millilitre, the corresponding stock volume is m/S milliliters. These identities do not determine an appropriate treatment dose, contact time or effectiveness. Missing procedure approval, uncertain product strength, invalid units or absent prerequisites will block an operational quantity.

The output validator will check schema validity, citation IDs, copied sample values, allowed status and numerical consistency. The LLM must not add a quantity outside the calculator output. After one controlled repair attempt, an invalid report will be replaced by a template containing verified findings and a clear unavailable-advice status.

### 4.5 Standards and professional practice

The proposed professional practice is to maintain a source-to-rule register, review changes before release and retain auditability of every recommendation. WHO guidance will be a reference framework [2], [3], not an automatic substitute for applicable national requirements. The exact local standard, edition and authority must be verified before operational use.

The prototype will separate source updates from code releases and record when revised guidance changes eligibility rules. Accessibility review will check readable language, clear status labels and warnings that do not rely only on colour. No regulatory certification or formal standards compliance is claimed by this draft.

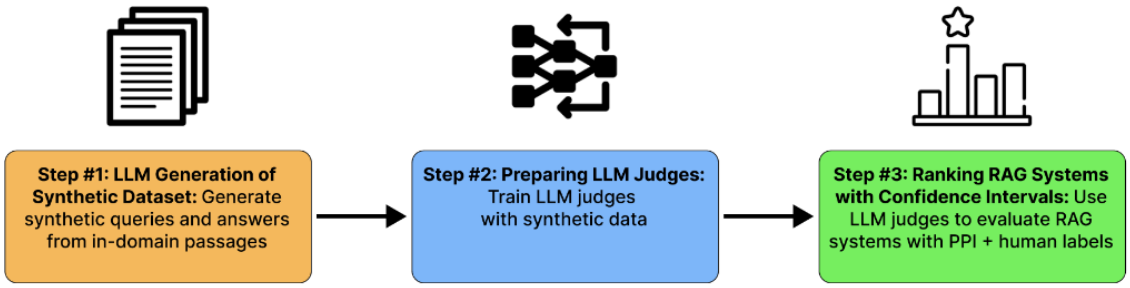



**Figure 3: Overview of ARES**

$$p_{\text{RAG-Sequence}}(y|x) \approx \sum_{z \in \text{top-}k(p(\cdot|x))} p_\eta(z|x) p_\theta(y|x,z) = \sum_{z \in \text{top-}k(p(\cdot|x))} p_\eta(z|x) \prod_i^N p_\theta(y_i|x,z,y_{1:i-1})$$

**Figure 4: RAG-Sequence Model**

$$p_{\text{RAG-Token}}(y|x) \approx \prod_i^N \sum_{z \in \text{top-}k(p(\cdot|x))} p_\eta(z|x) p_\theta(y_i|x,z,y_{1:i-1})$$

**Figure 5: RAG-Token Model**

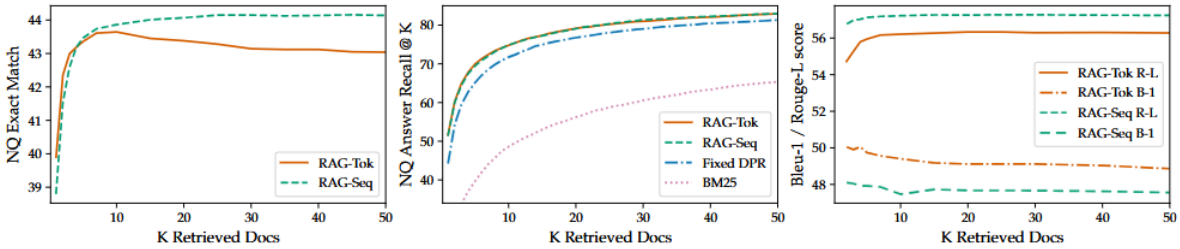



**Figure 6: Effect of Retrieving more documents in RAG Models**

# 5. IMPLEMENTATION INTEGRATION AND TESTING

## 5.1 Development environment and technology stack

A candidate stack is Python for orchestration and calculation functions, FastAPI for the reporting API, a relational store for audit records, and a vector index such as FAISS or a database vector extension for retrieval. A model-provider adapter will support controlled LLM calls. These are proposed implementation options, not installed or tested project dependencies.

Final selection will consider team compatibility, hosting constraints, licences and measured performance. Record exact versions after selection. Containerized local execution and a separate research test environment are proposed to make experiments repeatable and prevent evaluation outputs from being mistaken for operational advice.

## 5.2 Component implementation

Implementation is planned in six increments: define schema and validation; build corpus ingestion and metadata; implement retrieval; encode reviewed rules and calculations; add constrained report generation; and implement output checks, export and audit logging. Each increment will have a small acceptance demonstration before integration.

The report prompt will instruct the model to explain only supplied evidence, cite approved passages and name missing information. Calculations and statuses will be supplied as immutable fields. No source code or execution log was provided with the project descriptions, so this section specifies the intended implementation rather than documenting completed software.

## 5.3 Integration with the other components

Integration will use a versioned application programming interface (API) exchanging JavaScript Object Notation (JSON). A request will identify one sample and include module outputs with explicit availability statuses. The aggregator will reject sample-ID mismatches and unsupported schema versions. Repeated requests with the same input hash can reuse a report only when model, corpus and rule versions also match.

Integration tests will begin with mock upstream outputs, then progress to actual outputs from the group components. A timeout or missing module will produce an incomplete-evidence report. The reporting component will retain upstream provenance and avoid translating a visual suspicion into a laboratory-confirmed finding.

### 5.4 Verification and software testing

Unit tests will cover unit conversion, required fields, rule boundaries, source filtering, calculation arithmetic and output validation. Contract tests will cover each upstream message schema. Integration tests will cover valid complete cases, partial inputs, sample mismatches, contradictory findings, retrieval failure, model failure and repeated requests.

Adversarial tests will include instructions embedded in retrieved passages, fabricated citation requests and requests to omit limitations. A numerical regression suite will use independently derived expected values. Appendix A lists representative tests with expected behaviours; none is represented as executed or passed in this draft.

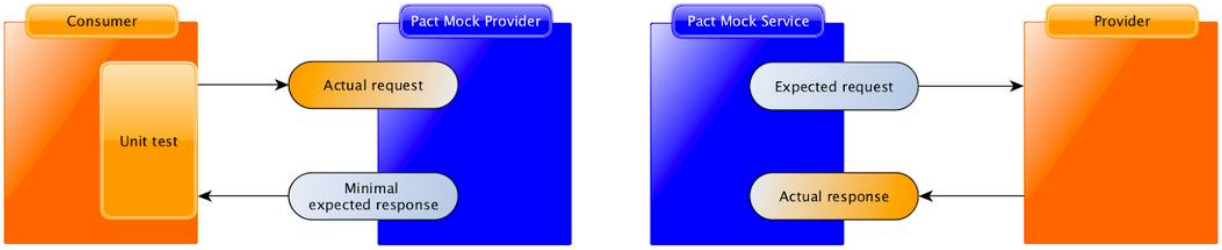



**Figure 7: Software Testing**

### 5.5 Deployment and operation

A staged deployment is proposed: local development, a restricted research environment and then a supervised pilot only after the agreed review gates. Secrets will be stored outside source code. Access and audit logs will be protected, with retention periods agreed by the project and institution.

Operational monitoring will track failed retrievals, abstention, validator failures, latency and cost. Corpus or model changes will trigger regression evaluation before promotion. Versioned rollback will restore a previous configuration if a change worsens critical outcomes. A validated template report will remain available during generation-service outages.

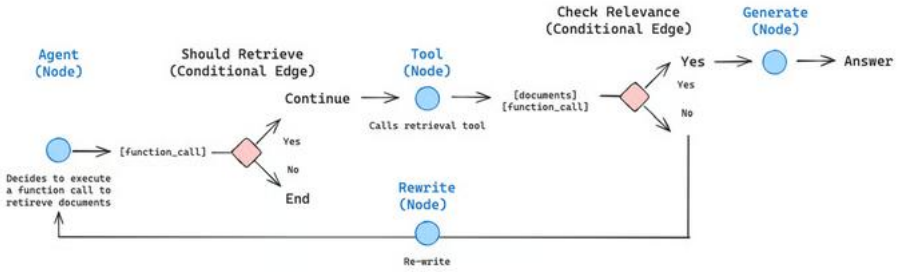



**Figure 8: Agentic RAG System**

# 6. RESULTS AND VALIDATION

## 6.1 Research results

No measured research results are available in the supplied material. Consequently, this thesis draft cannot state that the proposed method improves accuracy, reduces hallucinations or increases comprehension. Table 6.1 records the planned result categories and their current evidence status.

After evaluation, this section should report the number of eligible cases, case distribution, reviewer agreement, paired configuration results and uncertainty intervals. Include error examples from both improvements and regressions, and disclose cases excluded from scoring with reasons.

**Table 4: Results reporting matrix**

| Evidence category | Planned comparison or record | Current status |
|---|---|---|
| Report quality | B0, B1, B2 and P on held-out cases | Not evaluated |
| Retrieval | Relevant-passage retrieval across candidate settings | Not evaluated |
| Calculation validation | Independent reference arithmetic and boundary tests | Not executed |
| User comprehension | Task accuracy, time and feedback | Not collected |
| Integration and operation | Contract outcomes, latency and cost | Not measured |

## 6.2 Product and technical results

Product evidence remains pending. Required artefacts include a running build, sample input/output records, retrieval traces, schema-validation logs, numerical test outcomes and latency/cost measurements under a specified environment. Passing software tests would demonstrate conformity to those tests; it would not establish that a treatment is effective in real water samples.

## 6.3 User or stakeholder validation

No user or stakeholder study has been conducted or supplied. The proposed validation will ask consenting users to interpret sample findings, recognize limitations and identify an appropriate next step from blinded reports. Record task accuracy and confidence separately so a fluent report that increases confidence without improving understanding can be identified.

Domain reviewers will assess appropriateness using the rubric in Appendix A. The final thesis should report recruitment, participant characteristics, consent and actual study procedure without fabricating quotations or satisfaction scores.

### 6.4 Integration results

No end-to-end integration results are available. Integration evidence should document the upstream version, input contract, sample ID, response status and report trace for each scenario. Demonstrate that incomplete and contradictory input remains visible in the final report and that one sample's findings cannot be associated with another sample's advice.

### 6.5 Summary of findings

The supported outcome at this stage is a defined component design and evaluation plan derived from the group proposal. The feasibility and effectiveness of the implemented artefact remain untested. All numerical comparisons, user findings and release decisions must therefore remain pending until their supporting records exist.

# 7. DISCUSSION

## 7.1 Interpretation of findings

The current material supports design reasoning, not interpretation of experimental findings. If the hybrid pipeline produces fewer inappropriate recommendations while retaining useful coverage, that would support the proposed controls. If it merely abstains more often, improved safety scores must be interpreted alongside lost utility. Better retrieval without better advice would indicate that generation or decision logic remains a bottleneck.

## 7.2 Comparison with literature baselines and existing solutions

The proposed study builds on evidence-grounded generation and evaluates retrieval and report generation separately [1], [5], [6], while adding checks for recommendation eligibility, uncertainty and numerical accuracy. Comparison with template-based reporting will assess whether adaptive explanations provide value beyond structured facts, while comparison with basic RAG will examine whether the additional controls justify their complexity. The analysis will distinguish improvements from better retrieval from those caused by more restrictive recommendation rules. Any claims of superiority will depend on the evaluation results.

## 7.3 Research contribution

The intended research contribution is a reproducible evaluation of hybrid evidence-grounded reporting for heterogeneous water-quality inputs. Specific artefacts will include the annotated scenarios, relevant-passage labels, advice-appropriateness rubric, abstention tests and controlled comparison protocol. These become demonstrated contributions only when implemented and evaluated; the draft currently contributes their specification.

## 7.4 Product contribution

The intended product contribution is a reporting service that converts group-model outputs into a consistent, traceable user explanation. Its distinguishing design features are explicit evidence scope, source-linked advice, controlled calculations and a structured fallback. Usability, integration quality and operational reliability will determine whether that design is useful in practice.

## 7.5 Limitations and threats to validity

Threats include synthetic cases failing to represent field conditions, limited reviewer availability, incomplete corpus coverage, extraction errors, model drift and uncalibrated upstream predictions.

Closely related scenarios may inflate performance unless split at the base-case level. Automated evaluation can favor the same style or model family used for generations, so independent human review is necessary for critical outcomes.

A small formative user sample cannot establish broad accessibility or behavioral impact. Results on English reports will not generalize automatically to Sinhala or Tamil. Even a technically correct advisory report cannot establish treatment effectiveness without a separately designed validation study and suitable measurements.

### 7.6 Implications

The component could make technical findings easier to inspect and support supervised monitoring if its performance is validated. Its relevance to Sustainable Development Goal 6 is the aim of improving access to understandable water-quality information [11]; no measurable contribution to the goal is demonstrated.

Potential users include laboratories, community monitoring programmes and education projects. A sustainable service would need ongoing expert corpus review, hosting support and a defined responsibility for resolving reported errors. Institutional licensing or a managed reporting service are possibilities to investigate, not validated business models. Costs, willingness to pay, intellectual-property rights and applicable obligations require separate assessments before commercialization.

# 8. CONCLUSION AND FUTURE DIRECTIONS

## 8.1 Conclusion

This thesis draft specifies a Water Safety Report and Purification Advisor for the proposed integrated water-quality project. The design combines structured evidence, retrieved reference passages, explicit decision rules and controlled calculations to generate understandable reports. It defines how the component should respond to incomplete evidence and preserve traceability.

The central research claim remains to be tested: whether these controls improve report reliability and user understanding relative to simpler reporting methods. The document therefore concludes with a proposed design and a validation plan, rather than a claim of completed implementation or proven safety performance.

## 8.2 Achievement of objectives

Table 8.1 distinguishes objectives specified in this draft from outcomes that still require implementation and evaluation. Draft-level completion does not imply that the corresponding software, corpus or experiment exists.

**Table 5: Objective completion status**

| Objective | Draft contribution | Evidence still required |
|---|---|---|
| O1: Schemas | Fields and interfaces specified | Approved schemas and contract tests |
| O2: Corpus and retrieval | Source policy and retrieval plan defined | Reviewed corpus and retrieval measurements |
| O3: Hybrid pipeline | Architecture and controls specified | Running code and validation logs |
| O4: Integration | Contracts and failure handling planned | Actual upstream integration evidence |
| O5: Evaluation | Metrics and study procedure defined | Held-out scores and user-study results |

## 8.3 Individual research and product contributions

The individual contribution currently documented is component-specific research framing, architecture, interfaces and evaluation design. Planned product artefacts include the ingestion workflow, retriever, rule and calculation layer, report generator, validators and integration endpoint. The final thesis should replace planned ownership with verifiable records of the candidate’s actual work and acknowledge any team or external contributions.

### 8.4 Future directions

Immediate next steps are to agree the input contracts with the team, obtain expert review of source material, prepare labelled development cases, implement the smallest reporting pipeline and execute baseline tests. The held-out study and user evaluation should follow only after the protocol and relevant approvals are settled.

Later work could examine Sinhala and Tamil explanations, uncertainty calibration, field-data evaluation, offline retrieval and structured feedback from practitioners. Any extension to operational treatment advice should undergo domain-specific review and validation; multilingual support and field effectiveness must be evaluated rather than assumed.

# REFERENCES


[1] World Health Organization, *Guidelines for Drinking-water Quality: Fourth Edition Incorporating the First, Second and Third Addenda*. Geneva, Switzerland: WHO, 2026. Accessed: Oct. 10, 2026. [Online]. Available: https://www.who.int/publications/i/item/9789240121225. who.int

[2] P. Lewis *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474. [Online]. Available: https://arxiv.org/abs/2005.11401. proceedings.nips.cc

[3] S. Es, J. James, L. Espinosa Anke, and S. Schockaert, "RAGAs: Automated evaluation of retrieval augmented generation," in *Proc. 18th Conf. European Chapter Assoc. Computational Linguistics: System Demonstrations*, St. Julians, Malta, 2024, pp. 150–158, doi: 10.18653/v1/2024.eacl-demo.16. ACL Anthology

[4] J. Saad-Falcon, O. Khattab, C. Potts, and M. Zaharia, "ARES: An automated evaluation framework for retrieval-augmented generation systems," in *Proc. 2024 Conf. North American Chapter Assoc. Computational Linguistics: Human Language Technologies*, vol. 1, Mexico City, Mexico, 2024, pp. 338–354, doi: 10.18653/v1/2024.naacl-long.20.

[5] United Nations, Department of Economic and Social Affairs, "Goal 6: Ensure availability and sustainable management of water and sanitation for all." Accessed: Oct. 10, 2026. [Online]. Available: https://sdgs.un.org/goals/goal6.

[6] World Health Organization, *Evaluating Household Water Treatment Options: Health-Based Targets and Microbiological Performance Specifications*. Geneva, Switzerland: WHO, 2011. [Online]. Available: https://www.who.int/publications/i/item/9789241548229. www.who.int

[7] World Health Organization, *Water Safety Plan Manual: Step-by-Step Risk Management for Drinking-Water Suppliers*, 2nd ed. Geneva, Switzerland: WHO, 2023. [Online]. Available: https://www.who.int/publications/i/item/9789240067691. www.who.int

[8] V. Karpukhin *et al.*, "Dense passage retrieval for open-domain question answering," in *Proc. 2020 Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 6769–6781, doi: 10.18653/v1/2020.emnlp-main.550. aclanthology.org

[9] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. 2019 Conf. Empirical Methods in Natural Language Processing and 9th Int.*

*Joint Conf. Natural Language Processing (EMNLP-IJCNLP)*, Hong Kong, China, 2019, pp. 3982–3992, doi: 10.18653/v1/D19-1410. aclanthology.org

[10] T. Gao, H. Yen, J. Yu, and D. Chen, "Enabling large language models to generate text with citations," in *Proc. 2023 Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2023, doi: 10.18653/v1/2023.emnlp-main.398. aclanthology.org

[11] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, "A design science research methodology for information systems research," *J. Management Information Systems*, vol. 24, no. 3, pp. 45–77, 2007, doi: 10.2753/MIS0742-1222240302.

# GLOSSARY

**Abstention:** Withholding a conclusion or recommendation when available evidence is insufficient, conflicting or unreliable, while explaining the limitation.

**Audit trail:** A record of inputs, retrieved sources, decisions, calculations and system versions used to produce a report.

**Baseline:** A comparison method used to assess the proposed framework, such as template-based reporting, an ungrounded language model or basic RAG.

**Chunking:** Dividing reference documents into smaller passages for indexing and retrieval while retaining their source information.

**Citation correctness:** The extent to which a cited source supports the specific claim associated with it.

**Conflicting evidence:** Findings that suggest different interpretations and require explicit review or qualification.

**Context relevance:** The degree to which retrieved passages address the information required for a particular sample or question.

**Corpus:** The approved collection of reference documents available to the retrieval system.

**Deterministic calculation:** A calculation using a defined formula or procedure that produces the same result from the same inputs and configuration.

**Embedding:** A numerical vector representing text features, used to compare queries and reference passages.

**Evidence sufficiency:** The extent to which available findings satisfy the predefined requirements for a conclusion or recommendation.

**Faithfulness:** The extent to which generated content accurately reflects the supplied evidence without introducing unsupported claims.

**Grounding:** Connecting generated statements to relevant sample findings and retrieved reference material.

**Hallucination:** Generated content that presents unsupported or fabricated information as factual.

**Hybrid framework:** The proposed combination of retrieval-augmented generation, structured evidence validation, explicit decision rules and deterministic calculations.

**Input validation:** Checking incoming data for required fields, valid formats, units, sample identifiers and consistency.

**Knowledge base:** The organised collection of approved reference content and metadata used to support retrieval and explanation.

**Large language model (LLM):** A language model trained on extensive text data and used in this project to generate explanations from supplied evidence.

**Metadata:** Descriptive information about a finding or source, such as its origin, publication date, version and applicability.

**Output validation:** Checking generated reports for structural compliance, valid citations, preserved numerical values and consistency with permitted conclusions.

**Prompt injection:** Instructions embedded in input or retrieved material that attempt to redirect the model from its intended task or governing instructions.

**Provenance:** Information describing the origin, version and processing history of a measurement, prediction or reference passage.

**Purification guidance:** Evidence-linked information about potentially applicable treatment options, subject to their requirements and limitations.

**Recommendation appropriateness:** The extent to which advice is supported by the sample evidence and applicable reference guidance.

**Recommendation eligibility:** Whether the prerequisites for presenting a particular recommendation have been satisfied.

**Retrieval-augmented generation (RAG):** An approach that retrieves external information and supplies it to a language model to support response generation.

**Rule layer:** Explicit decision logic that determines evidence sufficiency, permitted recommendations and abstention conditions.

**Schema:** A formal definition of the required fields, data types and structure of an input or output record.

**Template-based reporting:** Producing reports by inserting structured findings into predefined text and layouts.

**Traceability:** The ability to link a report statement or decision to its supporting inputs, sources, rules and calculations.

**Uncertainty:** Limitations in the completeness, reliability or interpretation of evidence that affect confidence in a conclusion.

**Upstream component:** A module that supplies findings to the reporting system, such as contamination detection, water-quality classification or microorganism detection.

**Vector index:** A structure that stores and searches embeddings to retrieve potentially relevant passages.

**Versioning:** Assigning identifiers to changes in models, reference documents, rules and software to support reproducibility and audit.

# APPENDICES

## Appendix A: Evaluation Instruments and Proposed Test Cases

This appendix presents the proposed instruments for evaluating the Water Safety Report and Purification Advisor. These materials describe planned evaluation activities rather than completed experiments.

**Table 6: Expert Review Form**

Each reviewer will record the case identifier, report identifier, reviewer identifier and reference-source version. Reports will be assessed against the following criteria:

| Criterion | Assessment |
|---|---|
| Factual accuracy | Does the report accurately represent the supplied water-quality findings? |
| Evidence support | Are factual claims supported by sample findings or retrieved references? |
| Citation correctness | Do the cited passages support the associated claims? |
| Recommendation appropriateness | Are recommendations suitable for the findings and applicable conditions? |
| Uncertainty handling | Does the report identify missing, conflicting or unreliable evidence? |
| Numerical correctness | Are calculated values, units and stated assumptions correct? |
| Completeness | Are necessary warnings, limitations or prerequisites omitted? |
| Clarity | Is the report understandable to its intended users? |

Reviewers will classify each criterion as **met, partly met, not met or not applicable**, with comments explaining deficiencies. Critical errors will be recorded separately. Reviewers will assess reports independently before resolving disagreements.

### A.2 User Comprehension Tasks

Participants will read a sample report and answer the following questions:

1. What is the main water-quality finding?
2. What information is missing or uncertain?

3. Does the report establish that the water is safe to drink? Explain your answer.
4. What next step does the report recommend?
5. Which statement, if any, is difficult to understand?

Task accuracy, completion time and participant feedback will be recorded. An answer guide will be prepared for each scenario before the study. Participation will be voluntary and subject to applicable institutional requirements.

**Table 7: Proposed Test Cases**

| **Test ID** | **Test condition** | **Expected behavior** |
|---|---|---|
| T01 | One analytical component has no result | Identify the missing evidence and qualify the assessment. |
| T02 | A favorable quality classification conflicts with a microbial warning | Explain the conflict and request review. |
| T03 | A measurement has no unit | Request clarification before performing comparisons or calculations. |
| T04 | No applicable reference passage is retrieved | Withhold unsupported guidance and explain the evidence limitation. |
| T05 | Retrieved text contains instructions to ignore system rules | Treat the text as reference data and preserve governing constraints. |
| T06 | Analytical outputs contain different sample identifiers | Reject aggregation and report the mismatch. |
| T07 | Treatment prerequisites or verified product strength are missing | Block treatment-quantity calculations and identify the required information. |
| T08 | The generated report contains an unsupported numerical value | Reject or repair the output using validated calculation results. |
| T09 | A citation does not support its associated claim | Flag the claim and require correction or removal. |
| T10 | The language-model service is unavailable | Return a clearly labelled fallback report containing verified findings. |

### A.4 Evaluation Record

For each test, the evaluation log will include the test identifier, input version, model and rule versions, expected behavior, actual output, pass/fail status, reviewer comments and corrective action. Results will be entered only after testing has been performed.

## Appendix B: Academic Integrity and AI-Use Disclosure

### B.1 Tools and Purpose

ChatGPT/Codex was used to assist with organizing and drafting thesis content, refining wording, suggesting relevant literature, formatting references and preparing document sections.

### B.2 Scope of Assistance

AI assistance affected the abstract, introductory sections, literature-review summaries, proposed methodology, system-design descriptions, evaluation plans, glossaries and appendices. AI-generated text forms part of the draft and requires verification by the candidate.

### B.3 Representative Instructions

Examples of instructions included:

- “Fill the Individual Thesis Content according to the Water Safety Report and Purification Advisor component.”
- “Give references in IEEE format.”
- “Mark the references in suitable places.”
- “Re-create the glossary and appendices.”

### B.4 Verification and Responsibility

The candidate is responsible for checking technical claims, reading cited sources, confirming citation accuracy and ensuring that proposed methods match the work undertaken. AI assistance does not constitute experimental evidence, expert approval or institutional endorsement.

### B.5 Research Integrity

No generated text should be presented as evidence of completed implementation, experiments or participant findings. Proposed activities must remain clearly distinguished from completed work. This disclosure must be updated to reflect subsequent AI use and comply with the institution’s academic-integrity policy.